

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** Metro Ticket Booking System
- **Developed By :** Mathivanan S

2. Project Overview

- **Purpose:** The Metro Ticket Booking System is a ServiceNow-based application that enables passengers to book metro tickets online. The application automatically calculates the ticket fare, generates a QR code for digital ticket verification, and stores booking details in a custom ServiceNow table.

Features:

- Online Metro Ticket Booking
- Automatic Fare Calculation
- Dynamic QR Code Generation
- Digital Ticket Management
- Booking Confirmation
- Secure Record Storage
- User-Friendly Interface

3. Architecture

User Interface

- ServiceNow UI Page (Jelly)
- HTML
- CSS
- JavaScript

Application Logic

- Client Script
- Script Include

- GlideAjax
- GlideRecord

Database

- ServiceNow Custom Table (u_metro_ticket_booking).

4. Setup Instructions

Prerequisites

- ServiceNow Personal Developer Instance (PDI)
- System Administrator Role
- Application Scope
- Modern Web Browser

Installation

1. Create a ServiceNow application.
2. Create the custom table (u_metro_ticket_booking).
3. Create the Service Catalog Item or UI Page.
4. Configure Client Scripts.
5. Create the Script Include.
6. Configure GlideAjax.
7. Test the application.

5. Project Structure

Frontend

- UI Page
- HTML
- CSS

- JavaScript
- Jelly

Backend

- Client Scripts
- Script Include
- GlideAjax
- GlideRecord

Database

- u_metro_ticket_booking

6. Running the Application

1. Log in to the ServiceNow instance.
2. Open the Metro Ticket Booking application.
3. Enter passenger details.
4. Calculate fare automatically.
5. Generate the QR code.
6. Click Book Ticket.
7. Verify the booking record in the custom table.

7. Application Components

- Custom Application
- Custom Table
- UI Page
- Client Script
- Script Include

- GlideAjax
- QR Code Generator
- Flow Designer (Optional)

8. Security

- Role-Based Access Control (ACL)
- Secure GlideRecord Operations
- Input Validation
- Authenticated User Access

9. User Interface

- ☐ Metro Ticket Booking Form
- ☐ Fare Calculation
- ☐ QR Code Generation
- ☐ Booking Confirmation
- ☐ Booking Records.

10. Testing

Functional Testing

- Ticket Booking
- Fare Calculation
- QR Code Generation
- Database Record Creation

Performance Testing

- Response Time
- Booking Speed
- QR Generation Time

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11. Screenshots / Demo

- **Booking Form**

The screenshot shows the 'Book A Metro Ticket' form in the ServiceNow interface. The form includes the following fields and components:

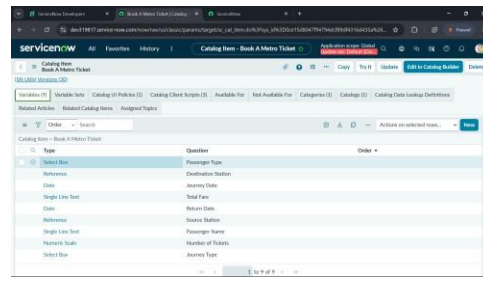
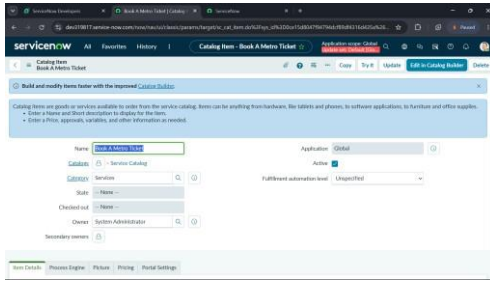
- Passenger Type:** A dropdown menu with 'Adult' selected.
- Destination Station:** A text input field with a search icon.
- Journey Date:** A date picker field.
- Total Fare:** A light blue bar representing the fare amount.
- Return Date:** A date picker field, marked with a red asterisk.
- Source Station:** A text input field with a search icon.
- Passenger Name:** A text input field.
- Order this Item:** A sidebar on the right containing:
 - Quantity:** A dropdown menu set to '1'.
 - Delivery time:** '0 Days'.
 - Order Now:** A blue button.
 - Add to Cart:** A button.
 - Shopping Cart:** A section labeled 'Empty'.

- **QR Code**

The screenshot shows the 'Metro Ticket QR Code' page in the ServiceNow interface. The page displays the following information:

- Destination Station:** A dropdown menu with 'Select' chosen.
- Journey Date:** A date input field showing 'dd-mm-yyyy'.
- Passenger Type:** A dropdown menu with 'Adult' selected.
- No. of Tickets:** A text input field with the value '1'.
- Total Fare:** A text input field with the value '50'.
- Metro Ticket QR Code:** A large QR code displayed below the form fields.

- **Service Catalog**



• Booking Record

ServiceNow Developers x Metro Ticket Bookings | Service: x ServiceNow

dev319817.service-now.com/now/nav/ui/classic/params/target/u_metro_ticket_booking_list.do%3Fsysparm_userpref_module%3Dc391...

Application scope: Global
Update set: Default [Cio]

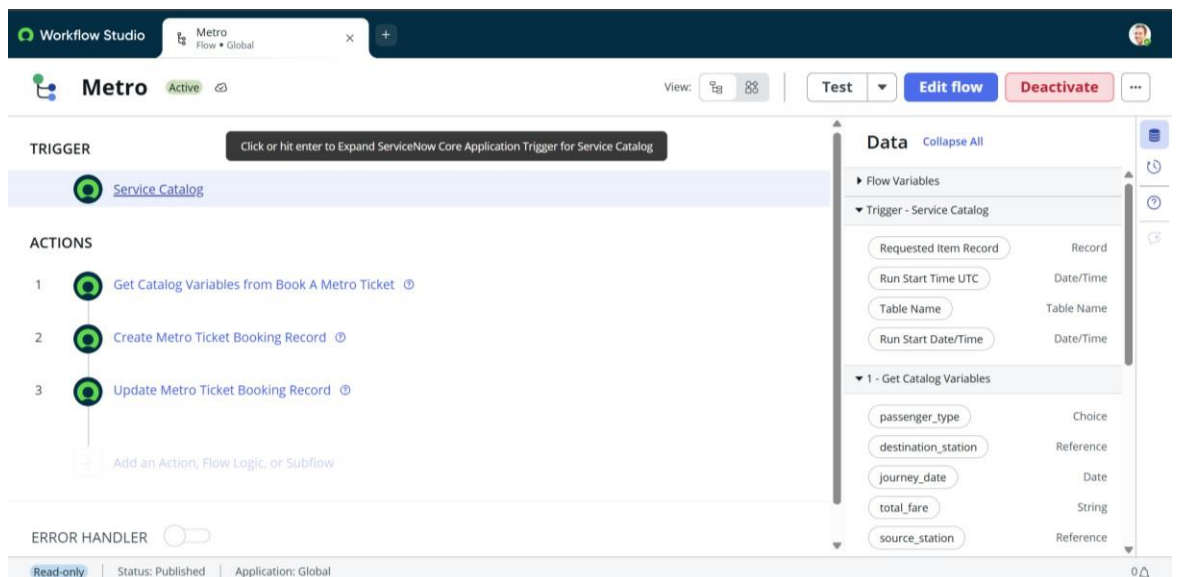
Metro Ticket Bookings Booking Status Search

Actions on selected rows... New

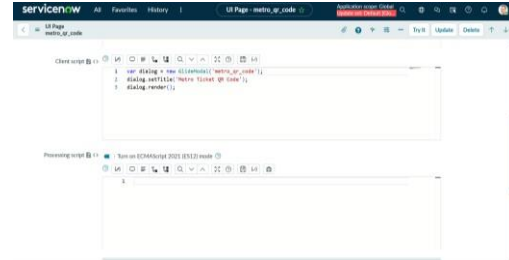
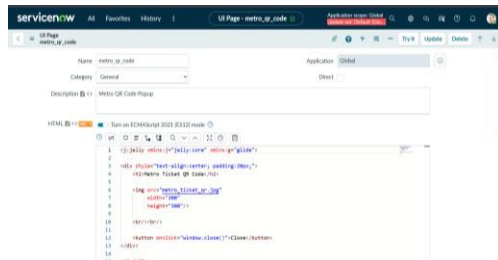
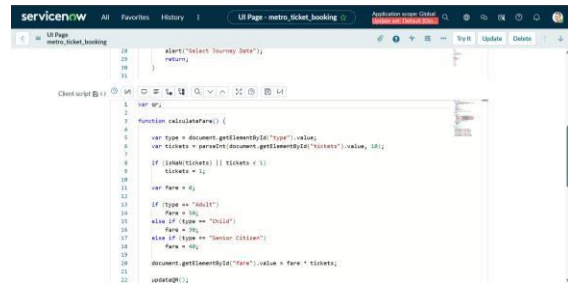
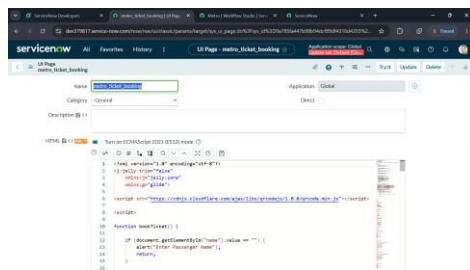
Booking Status	Destination Station	Journey Date	Mobile Number	Number of Tickets	Passenger Name	Passenger Type	Payment Status
Search	Search	Search	Search	Search	Search	Search	Search
(empty)	(empty)	2026-07-09		5	Ragul	Senior Citizen	
(empty)	(empty)	2026-07-10		5	dass	Adult	
Booked	Airport	2026-06-26		4	Roja	adult	Pending
Booked	Airport	2026-06-30		3	Siva Mounith	adult	Pending
Booked	Airport	2026-07-09		3	ramasamy	child	Pending
Booked	Airport	2026-07-09		4	Saal	adult	Pending
Booked	Airport	2026-07-10		4	Kaviya	adult	Pending
Booked	Airport	2026-07-09		5	Phillips	senior Citizen	Pending
Booked	Airport	2026-07-10		0	Mano	child	Pending
Booked	Airport	2026-07-10		2	Ramu	adult	Pending
(empty)	(empty)	2026-06-30		5	Ravi	Senior Citizen	

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• Flow Designer



• UI Page



12. Known Issues

- Requires an active internet connection.
- QR code library must be available.
- Depends on ServiceNow instance availability.

13. Future Enhancements

- Online Payment Gateway Integration
- Email and SMS Notifications
- Mobile Application Support
- Metro Smart Card Recharge
- Live Metro Timetable
- Booking History Dashboard
- Multi-Language Support
- QR Code Validation at Metro Entry Gates